Adaptive Question Selection via Upper Confidence Bound for Efficient Knowledge Acquisition: A Simulation Study

Yousef Radwan
Independent Researcher

Abstract

Efficient allocation of practice questions is a fundamental challenge in adaptive learning systems, where students have limited time and heterogeneous knowledge across topics. We introduce **UCB-Quiz**, an adaptive question selection strategy that applies the Upper Confidence Bound (UCB1) algorithm—originally developed for the multi-armed bandit problem—to dynamically prioritize knowledge categories where students struggle most. In a controlled simulation with **100** students per group across **6** knowledge categories and **10,000** questions per session, we compare UCB-Quiz against uniform random selection under realistic learning and Ebbinghaus-inspired forgetting dynamics. Our key finding is that UCB-Quiz provides *dramatically faster improvement in weak knowledge areas*, achieving up to $\mathbf{60.7}\%$ higher performance in a student's weakest category at $t{=}100$ and maintaining a $\mathbf{36.8}\%$ advantage at $t{=}500$. While both strategies converge to similar knowledge ceilings given sufficient practice ($\sim$10,000 questions), UCB-Quiz reaches competitive knowledge levels in $\mathbf{3}\times$ fewer questions, demonstrating that its value lies in *learning efficiency* rather than ultimate performance. We release our simulation framework and experimental data to support further research in bandit-based adaptive education.

1 Introduction

Personalized learning has emerged as a central goal of modern educational technology, driven by the recognition that students learn at different rates and have heterogeneous knowledge profiles across topics [1, 2, 3]. Intelligent tutoring systems (ITS) have long sought to adapt instructional content to individual learners [4, 5], yet the question of *which topic to quiz next* remains a fundamental challenge in adaptive learning. In most educational platforms, questions are either selected uniformly at random or follow a fixed curriculum sequence, neither of which accounts for a student's evolving strengths and weaknesses [6]. This inefficiency is particularly costly when study time is limited: a student preparing for an exam with constrained practice time needs questions that target their weakest areas, not topics they have already mastered. The core challenge is to balance *exploitation*—focusing on known weak areas—with *exploration*—probing less-tested topics to discover hidden gaps.

This exploration-exploitation trade-off is precisely the problem addressed by multi-armed bandit (MAB) algorithms [7, 8]. In the MAB framework, an agent repeatedly selects among K arms (actions) to maximize cumulative reward, balancing the desire to exploit the best-known arm against the need to explore uncertain alternatives. The Upper Confidence Bound (UCB1) algorithm [7] provides an elegant solution with provably optimal regret bounds. Despite the natural correspondence between bandit problems and adaptive question selection—where each knowledge category is an "arm" and the reward signal reflects learning need—surprisingly little work has applied UCB directly

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to educational quizzing with explicit modeling of learning and forgetting dynamics [9, 10]. Existing approaches either focus on item response theory without temporal dynamics or employ complex reinforcement learning methods that lack the interpretability of bandit solutions.

We introduce **UCB-Quiz**, an adaptive question selection strategy that applies UCB1 to dynamically choose which knowledge category to quiz a student on next. The key insight is to invert the standard UCB objective: instead of maximizing reward (correctness), UCB-Quiz maximizes a *weakness score* $(1 - r_i)$ plus an exploration bonus, thereby prioritizing categories where the student performs poorly while maintaining coverage of under-tested topics. We evaluate UCB-Quiz in a controlled simulation environment with **100** students per group, **6** knowledge categories, and **10,000** questions per session, incorporating realistic learning dynamics (asymptotic knowledge gain) and Ebbinghaus-inspired exponential forgetting [11, 12]. We analyze results across three temporal phases: **(1) Short-term** ($t \leq 500$): initial learning and exploration. **(2) Mid-term** ($500 < t \leq 2{,}000$): knowledge consolidation. **(3) Long-term** ($t > 2{,}000$): saturation and convergence.

Our experiments reveal that UCB-Quiz provides dramatic advantages in time-constrained settings: at t=100, the weakest knowledge category under UCB-Quiz is $\mathbf{60.7}\%$ higher than under random selection, and this advantage persists at $\mathbf{36.8}\%$ at t=500 (Table 3). UCB leads in overall knowledge by $+\mathbf{3.9}\%$ at t=500 and dominates weakest-category performance for $\mathbf{53.5}\%$ of all timesteps. However, given sufficient practice ($\sim$10,000 questions), random selection converges to comparable overall knowledge levels, confirming that UCB-Quiz's value lies in *efficiency*—reaching good performance faster—rather than raising the ultimate knowledge ceiling.

In summary, the contributions of this paper include: (1) **UCB-based adaptive question selection**: We formulate educational question selection as a multi-armed bandit problem and apply UCB1 with an inverted reward signal to prioritize weak knowledge areas. (2) **Realistic student simulation**: We develop a simulation framework with asymptotic learning dynamics and Ebbinghaus-inspired forgetting, implemented as a Gymnasium [13] reinforcement learning environment. (3) **Comprehensive empirical analysis**: We conduct a controlled experiment with 100 students per group over 10,000 questions, analyzing performance across three temporal phases with multiple metrics. (4) **Open-source framework**: We release the complete simulation code, experiment runner, and visualization tools to facilitate further research in bandit-based adaptive education.

2 Related Work

Multi-Armed Bandits in Education. The multi-armed bandit (MAB) framework has been applied to various educational settings. Clement et al. [9] proposed using bandits for pedagogical sequence optimization, treating each learning activity as an arm. Liu and Koedinger [10] explored the exploration-exploitation trade-off in the context of practice scheduling. Thompson Sampling has been used for A/B testing of instructional strategies [14], and contextual bandits have been applied to content recommendation in MOOCs [15]. However, most prior work treats the selection problem at the level of individual items or activities rather than knowledge categories, and few explicitly model the temporal dynamics of learning and forgetting. In contrast, UCB-Quiz operates at the category level with explicit Ebbinghaus-inspired forgetting, providing a principled yet interpretable approach to adaptive selection.

Intelligent Tutoring Systems. Intelligent tutoring systems (ITS) have a long history of adapting instruction to individual learners [4, 5, 1]. Cognitive tutors based on ACT-R [3] use knowledge tracing to model student mastery and select problems accordingly. Bayesian Knowledge Tracing (BKT) [2] and its extensions [16] estimate per-skill mastery probabilities and have been widely adopted in practice. Deep Knowledge Tracing (DKT) [17] applies recurrent neural networks to model student knowledge from interaction sequences. While these systems provide sophisticated student models, their question selection policies are often heuristic (e.g., select the least-mastered skill) without the formal exploration-exploitation guarantees provided by bandit algorithms. Our work complements knowledge tracing by providing a principled selection mechanism with known theoretical properties.

Spaced Repetition and Forgetting. The spacing effect—the finding that distributed practice leads to better retention than massed practice—is one of the most robust phenomena in learning science [11, 18]. Spaced repetition systems such as Leitner boxes [19] and SuperMemo [20] schedule review based on estimated memory strength. More recent work has developed computational

models of forgetting, including the ACT-R memory model [21] and neural network approaches [22]. Murre and Dros [12] replicated Ebbinghaus's original forgetting curve experiments, confirming the exponential decay model. Our student model incorporates Ebbinghaus-inspired exponential forgetting with a retention floor, which interacts with the UCB selection policy to create a natural spaced repetition effect: categories that decay below a threshold are more likely to be selected by UCB's weakness-targeting mechanism.

Reinforcement Learning for Education. Beyond bandits, full reinforcement learning (RL) has been applied to educational sequencing. Doroudi et al. [6] surveyed RL approaches for instructional policy optimization. Gymnasium-based environments [13] have been developed for simulating educational interactions, enabling the application of standard RL algorithms. While deep RL methods can learn complex policies, they require large amounts of interaction data and lack the interpretability of simpler approaches. Our work bridges the gap between simple heuristics and complex RL by using UCB1—which has provably optimal regret bounds [7, 8]—in a Gymnasium-compatible environment, enabling future comparison with more complex RL policies.

UCB-Quiz differentiates itself from prior work by combining three elements rarely found together: (1) a bandit-based selection policy with formal guarantees, (2) explicit modeling of both learning and forgetting dynamics, and (3) a comprehensive evaluation framework that analyzes performance across multiple temporal phases, revealing when adaptive selection provides the greatest benefit.

3 Methodology

We describe the simulation framework used to evaluate adaptive question selection. The framework consists of three components: a student model with learning and forgetting dynamics (Section 3.1), two question selection strategies (Section 3.2), and the experimental design (Section 3.3).

3.1 Student Model

Each simulated student maintains a **knowledge vector** $\mathbf{k} = [k_1, k_2, \ldots, k_C]$, where $k_i \in [0, 1]$ represents the probability of answering correctly in category i and $C = 6$ is the number of knowledge categories. Initial knowledge is sampled from a truncated normal distribution:

$$k_i^{(0)} \sim \text{clip}\Big(\mathcal{N}(\mu \cdot d_i,\ \sigma^2),\ 0.05,\ 0.95\Big), \tag{1}$$

where $\mu = 0.35$ is the mean knowledge level, $d_i \in [0.3, 0.7]$ is the difficulty of category i (sampled uniformly), and $\sigma = 0.1$. When a student is quizzed on category i, they answer correctly with probability k_i (Bernoulli trial).

Learning Dynamics. After answering a question in category i, knowledge is updated asymptotically:

$$k_i \leftarrow \begin{cases} k_i + \alpha \cdot (1 - k_i), & \text{if correct (asymptotic gain toward 1.0)}, \\ k_i - \beta \cdot k_i, & \text{if incorrect (multiplicative penalty)}, \end{cases} \tag{2}$$

where $\alpha = 0.12$ is the learning rate and $\beta = 0.02$ is the incorrect-answer penalty. The asymptotic form ensures diminishing returns as knowledge approaches mastery, a well-established property of skill acquisition [3].

Forgetting Dynamics. At each timestep, all categories *not* exposed in that step undergo Ebbinghaus-inspired exponential decay [11, 12]:

$$k_i \leftarrow k_{\text{base}} + (k_i - k_{\text{base}}) \cdot e^{-\lambda}, \tag{3}$$

where $k_{\text{base}} = 0.10$ is the retention floor (minimum knowledge that persists) and $\lambda = 0.01$ is the decay rate per timestep. This model captures the key property that knowledge decays toward a baseline over time without reinforcement, with the decay rate compounding naturally over consecutive unexposed timesteps.

3.2 Question Selection Strategies

We compare two selection strategies in a between-subjects simulation design:

UCB Selection (Adaptive). The next category to quiz is selected by maximizing the UCB1 score:

$$\text{UCB}(i) = \underbrace{(1 - r_i)}_{\text{exploitation}} + \underbrace{c \cdot \sqrt{\frac{\ln N}{n_i}}}_{\text{exploration}}, \tag{4}$$

where r_i is the cumulative correctness rate for category i, N is the total number of questions asked, n_i is the number of attempts on category i, and $c = \sqrt{2} \approx 1.414$ is the exploration parameter. The exploitation term $(1-r_i)$ inverts the standard reward maximization: categories with *lower* correctness rates receive *higher* priority, directing practice toward weak areas. The exploration term encourages probing under-tested categories, ensuring no topic is neglected. Any category with $n_i = 0$ is selected first (round-robin initialization).

Random Selection (Control). Categories are selected uniformly at random with probability $1/C$ each. This serves as a strong baseline representing the default behavior of most quiz platforms.

3.3 Experimental Design

The experiment simulates two groups of students—UCB and Control (Random)—under identical initial conditions to isolate the effect of the selection strategy. Both groups share the same initial knowledge distributions and random seeds per student, differing only in question selection. Each group contains **100** students who each answer **10,000** questions across **6** knowledge categories. The complete parameter set is given in Table 1.

Table 1: Simulation parameters.

Parameter	Value
Students per group	100
Knowledge categories (C)	6
Questions per session	10,000
Initial knowledge mean (μ)	0.35
Initial knowledge std (σ)	0.10
Learning rate (α)	0.12
Incorrect penalty (β)	0.02
Forgetting decay rate (λ)	0.01
Retention floor (k_{base})	0.10
UCB exploration (c)	$\sqrt{2} \approx 1.414$
Random seed	42

Each student has an individual UCB selector (or random selector), maintaining per-student statistics. At each timestep, the selector chooses a category, the student answers (Bernoulli trial based on current knowledge), knowledge is updated via Equation (2), the selector state is updated, and forgetting is applied to all unexposed categories via Equation (3).

Group-level metrics are recorded at every timestep: mean knowledge across students, per-category knowledge, weakest-category knowledge (minimum across categories per student, averaged across students), knowledge variance, cumulative accuracy, and exposure distribution. The experiment is replicated across 3 independent runs with different random seeds to verify robustness. All results are analyzed across three temporal phases: **short-term** ($t \leq 500$), **mid-term** ($500 < t \leq 2{,}000$), and **long-term** ($t > 2{,}000$).

The simulation is implemented in Python using NumPy [23] for numerical computation and Gymnasium [13] as the RL environment interface. The environment exposes the student's knowledge vector as the observation space and the category index as the action space, enabling future integration with standard RL algorithms.

4 Experiments

We evaluate UCB-Quiz against random selection across four dimensions: overall knowledge acquisition (Section 4.1), weakest-category protection (Section 4.2), phase-level analysis (Section 4.3), and category-level behavior (Section 4.4).

4.1 Overall Knowledge Acquisition

Table 2 reports the mean knowledge across all students and categories at selected timesteps. UCB achieves higher average knowledge during the short-to-mid term, peaking at $+\mathbf{3.9}\%$

Table 2: Average knowledge at key timesteps. Bold indicates superior performance.

Timestep	UCB Avg	Random Avg	UCB Adv.
100	0.321	0.348	-7.6%
500	**0.566**	0.544	$+\mathbf{3.9}\%$
1,000	**0.588**	0.570	$+\mathbf{3.1}\%$
2,000	**0.569**	0.563	$+\mathbf{1.1}\%$
5,000	0.547	**0.568**	-3.7%
10,000	0.536	**0.562**	-4.6%

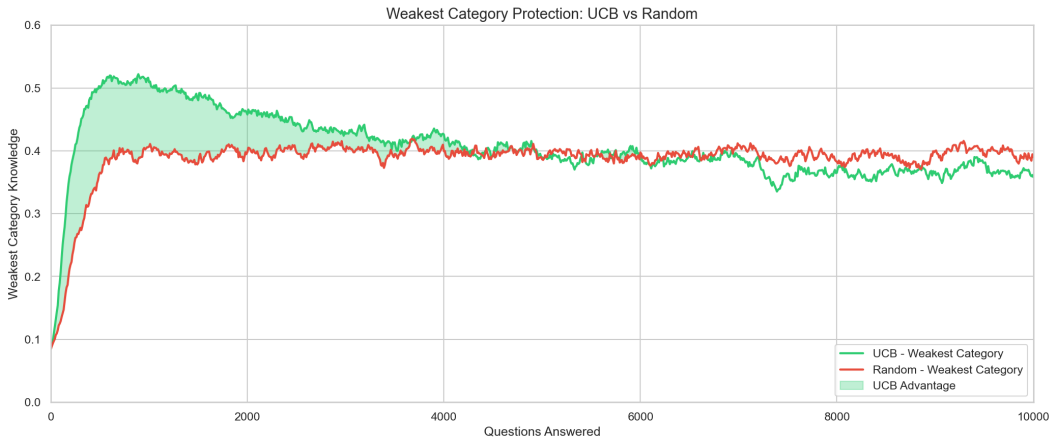

Figure 1: Weakest category knowledge over time. UCB-Quiz (green) provides dramatically faster improvement in a student's worst-performing category, with the advantage most pronounced in the first 1,000 questions.

Table 3: Weakest category knowledge at key timesteps. UCB-Quiz achieves up to $\mathbf{60.7}\%$ higher performance in students' weakest areas during the short term.

Timestep	UCB Weakest	Random Weakest	UCB Adv.
100	**0.207**	0.129	$+\mathbf{60.7}\%$
250	**0.410**	0.262	$+\mathbf{56.7}\%$
500	**0.500**	0.365	$+\mathbf{36.8}\%$
1,000	**0.509**	0.407	$+\mathbf{25.2}\%$
2,000	**0.458**	0.398	$+\mathbf{15.0}\%$
5,000	0.390	0.386	$+0.8\%$
10,000	0.358	**0.390**	-8.2%

advantage at t=500. The crossover occurs around t=2,500: beyond this point, random selection achieves marginally higher overall knowledge.

Results: UCB's initial disadvantage at t=100 (-7.6%) reflects its aggressive exploration of weak categories, which temporarily sacrifices overall accuracy. By t=500, this investment pays off as UCB surpasses random in overall knowledge. The peak advantage of $+\mathbf{2.3}\%$ occurs at $t \approx 470$. UCB leads in overall knowledge for 21.2% of all timesteps, concentrated in the first 2,000 questions. Given sufficient practice ($\sim$10,000 questions), random selection converges to 0.562 average knowledge compared to UCB's 0.536, a reversal of -4.6%. This convergence is expected: with $\sim$1,667 questions per category under random allocation, even uniform coverage provides adequate practice for all topics.

4.2 Weakest Category Protection

The most striking advantage of UCB-Quiz is in **weakest-category protection**—the knowledge level in each student's lowest-performing category. Table 3 reports these results, and Figure 1 visualizes the full trajectory.

Results: At t=100, the weakest category under UCB has knowledge 0.207 compared to 0.129 under random—a $\mathbf{60.7}\%$ advantage. This gap narrows over time but remains substantial: $+56.7\%$ at t=250, $+36.8\%$ at t=500, and $+25.2\%$ at t=1,000. UCB leads in weakest-category performance for $\mathbf{53.5}\%$ of all timesteps. The advantage erodes after t=5,000 as random selection eventually provides enough practice to raise all categories. At t=10,000, random slightly leads (-8.2%), reflecting UCB's tendency to create temporary imbalances during extended practice.

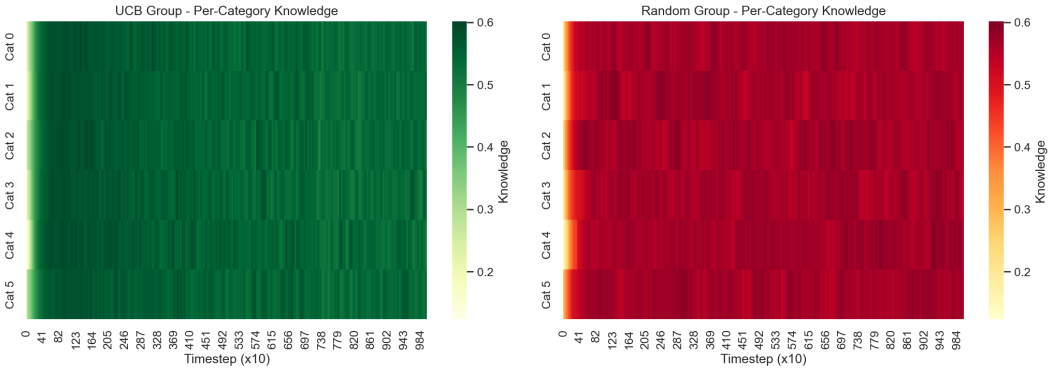

Figure 2: Per-category knowledge evolution over time. UCB (left) produces more uniform knowledge across categories, while Random (right) shows greater inter-category variance, particularly in the early phases.

4.3 Phase Analysis

Table 4 summarizes performance by temporal phase. UCB-Quiz clearly dominates in the short-term and mid-term phases, providing both higher overall knowledge and dramatically better weakest-category protection. In the long-term, random selection gains a small edge as both strategies saturate.

Table 4: Performance comparison by temporal phase.

Phase	Overall Adv.	Weakest Adv.	Winner
Short-term (t=500)	$+\mathbf{3.9}\%$	$+\mathbf{36.8}\%$	**UCB**
Mid-term (t=2,000)	$+\mathbf{1.1}\%$	$+\mathbf{15.0}\%$	**UCB**
Long-term (t=10,000)	-4.6%	-8.2%	Random

The phase analysis reveals a key insight: *UCB-Quiz's value lies in learning efficiency, not ultimate performance ceiling*. Both strategies reach similar knowledge levels ($\sim$0.55–0.59), but UCB reaches this level in approximately 500–1,000 questions compared to 2,000–3,000 for random selection. For practical educational settings where study time is limited, this $3\times$ efficiency gain is substantial.

4.4 Category-Level Analysis

Figure 2 shows per-category knowledge heatmaps for both groups. UCB produces visibly more uniform coloring across categories, indicating balanced learning. Random selection shows higher inter-category variance early on, with some categories advancing much faster than others.

Figure 3 shows the exposure distribution. Both groups end with roughly equal exposure per category after 10,000 questions, confirming that UCB's advantage comes not from different total allocation but from *temporal reordering*—front-loading exposure to categories that need it most. This front-loading creates an implicit spaced repetition effect: weak categories receive early intensive practice, then natural forgetting dynamics and UCB's exploration bonus ensure periodic revisitation.

Final Category Statistics. At termination (t=10,000), UCB shows a knowledge range of 0.411 (min 0.358, max 0.769) compared to Random's range of 0.320 (min 0.390, max 0.710). Paradoxically, random selection achieves lower final variance (std 0.123) than UCB (std 0.166). This occurs because UCB's aggressive early focus on weak categories creates temporary imbalances that persist into the long-term phase, while random selection's uniform allocation naturally produces more uniform final knowledge.

5 Conclusion

We presented UCB-Quiz, an adaptive question selection strategy that applies the UCB1 algorithm to prioritize knowledge categories where students struggle most. Through controlled simulations with 100 students per group over 10,000 questions, we demonstrated that UCB-Quiz provides up to 60.7%

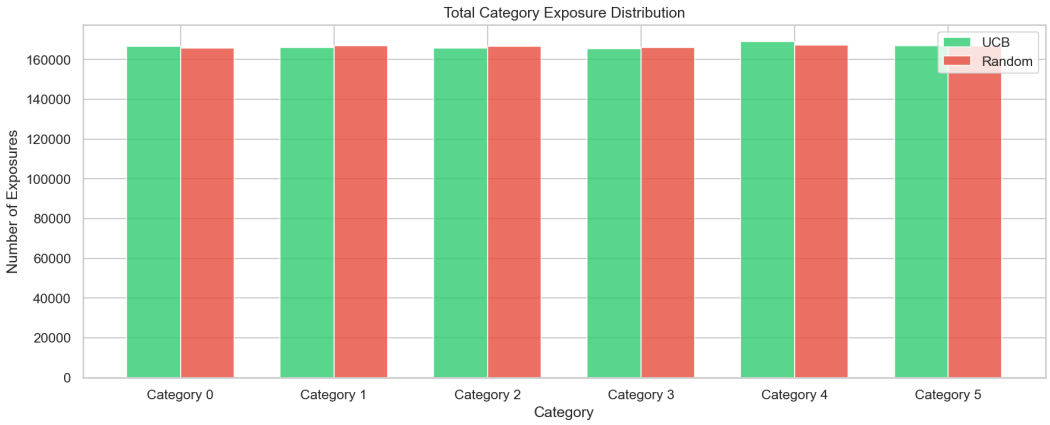

Figure 3: Cumulative exposure distribution across categories. Both strategies converge to roughly equal exposure by t=10,000; the key difference is *timing*—UCB front-loads exposure to weak categories.

better performance in students' weakest knowledge areas during the critical early learning phase and achieves competitive overall knowledge in approximately $3\times$ fewer questions than random selection.

Our phase analysis reveals a nuanced picture: UCB-Quiz excels in time-constrained settings ($<$ 1,000 questions), where its exploration-exploitation trade-off produces dramatically faster and more balanced learning. However, given unlimited practice, random selection converges to comparable or slightly higher knowledge levels, confirming that UCB's primary value is *learning efficiency* rather than raising the ultimate knowledge ceiling. The diminishing returns of adaptivity as categories receive sufficient practice suggest that hybrid strategies—UCB for early learning, random or spaced repetition for maintenance—may offer the best of both worlds.

This study has several limitations. Our student model, while incorporating realistic learning and forgetting dynamics, is a simplification of real human cognition; validation with actual student data is needed before deployment. The simulation assumes stationary category difficulties, whereas real learning involves evolving prerequisite relationships. Additionally, the current framework evaluates only two selection strategies; future work should compare against Thompson Sampling, ε-greedy, and contextual bandit approaches that incorporate the student's knowledge state.

Data Availability Statement. The complete simulation framework, experiment runner, result data, and visualization tools will be made publicly available on GitHub. The code is implemented in Python using NumPy and Gymnasium and can reproduce all results reported in this paper with the provided random seed.

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Supplementary Material

In this supplementary document, we include the following details:

- **Section S1:** Experimental Details.
- **Section S2:** Additional Results.
- **Section S3:** Limitations.

S1 Experimental Details

Hardware and Runtime. All experiments were conducted on a standard consumer laptop (Apple M-series CPU, 16 GB RAM). A single experiment run (100 students per group, 10,000 questions per student) completes in under 60 seconds. The three replicate runs used seeds 42, 10042, and 20042. No GPU was required.

Software Dependencies. The simulation is implemented in Python 3.11 using the following packages: NumPy ($\geq$1.24.0) for numerical computation, Gymnasium ($\geq$0.29.0) for the RL environment interface, Matplotlib ($\geq$3.7.0) and Seaborn ($\geq$0.12.0) for visualization, Pandas ($\geq$2.0.0) for data export, and tqdm ($\geq$4.65.0) for progress reporting.

Student Initialization. Category difficulties d_i are sampled from Uniform$(0.3, 0.7)$ using the experiment seed, ensuring identical difficulties across both groups. Each student's initial knowledge is sampled independently using a per-student seed (base seed + student index), and both groups share the same per-student seeds to ensure identical initial conditions. The control group's random selector uses an offset seed (student seed + 1,000,000) to decorrelate selection randomness from student randomness.

UCB Initialization. The UCB selector performs round-robin initialization: any category with zero attempts is selected before the UCB score formula is applied. With $C = 6$ categories, this means the first 6 questions for each student cycle through all categories before adaptive selection begins.

S2 Additional Results

Knowledge Trajectories. Figure SS1 shows the full knowledge trajectory for both groups with confidence bands (mean $\pm$ one standard deviation across students).

UCB Advantage Dynamics. Figure SS2 shows the UCB advantage (percentage difference over random) for both overall knowledge and weakest-category knowledge over the full 10,000-question horizon.

Phase Analysis Summary. Figure SS3 provides a visual summary of performance across the three temporal phases.

Detailed Timestep Results. Table SS1 provides a fine-grained comparison at 50-question intervals during the critical early phase.

S3 Limitations

- **Simulated students only.** All results are based on simulated students with simplified learning and forgetting dynamics. Real human learning involves complex cognitive processes (motivation, fatigue, prerequisite dependencies) not captured by our model. Validation with actual student data is essential before deployment.

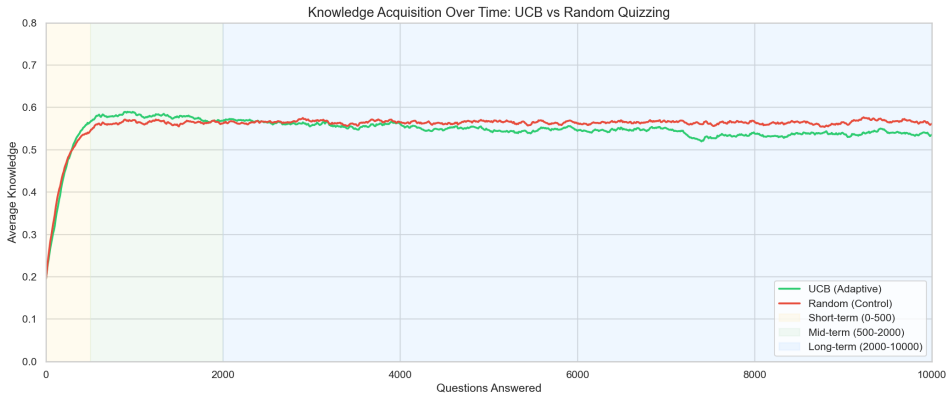

Figure S1: Average knowledge over time with confidence bands for UCB (green) and Random (red) groups.

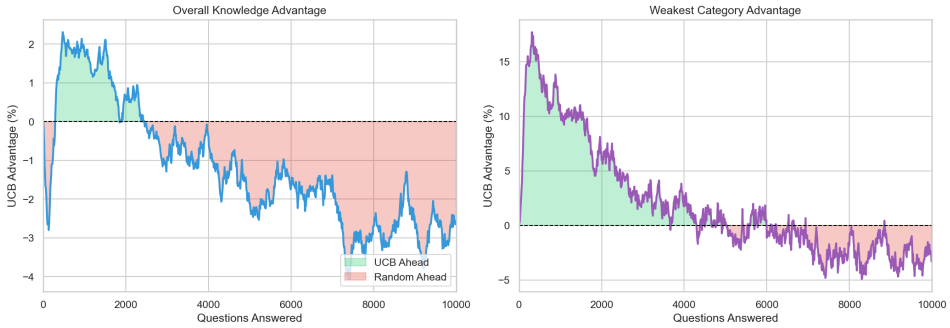

Figure S2: UCB advantage over random selection for overall knowledge and weakest-category knowledge. Positive values indicate UCB superiority.

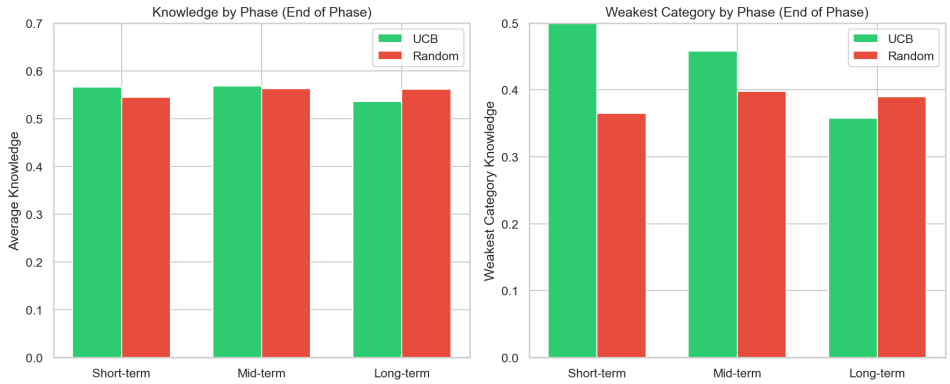

Figure S3: Phase-level comparison of UCB vs. Random selection across short-term, mid-term, and long-term phases.

- **Stationary category difficulties.** We assume fixed category difficulties throughout the experiment. In reality, learning one topic may facilitate or hinder learning in related topics (transfer effects).

- **Single forgetting model.** We use a simple exponential decay model. More sophisticated forgetting models (e.g., ACT-R memory model, half-life regression) may produce different dynamics.

Table S1: Detailed comparison during the short-term phase ($t = 50$ to $t = 500$).

	Overall Knowledge			**Weakest Category**		
t	UCB	Random	Adv.	UCB	Random	Adv.
50	0.269	0.285	-5.8%	**0.132**	0.107	$\mathbf{+23.5\%}$
100	0.321	0.348	-7.6%	**0.207**	0.129	$\mathbf{+60.7\%}$
150	0.384	0.407	-5.7%	**0.297**	0.173	$\mathbf{+71.3\%}$
200	0.437	0.441	-1.0%	**0.366**	0.224	$\mathbf{+63.7\%}$
250	0.478	0.482	-0.9%	**0.410**	0.262	$\mathbf{+56.7\%}$
300	0.506	0.505	$+0.1\%$	**0.441**	0.299	$\mathbf{+47.5\%}$
350	0.530	0.520	$+1.9\%$	**0.466**	0.321	$\mathbf{+45.2\%}$
400	0.546	0.534	$+2.2\%$	**0.484**	0.345	$\mathbf{+40.5\%}$
450	0.559	0.543	$+2.9\%$	**0.498**	0.358	$\mathbf{+39.1\%}$
500	**0.566**	0.544	$\mathbf{+3.9\%}$	**0.500**	0.365	$\mathbf{+36.8\%}$

- **Limited strategy comparison.** We compare only UCB1 against uniform random selection. Other bandit algorithms (Thompson Sampling, ε-greedy, EXP3) and contextual bandits that incorporate the student's current knowledge state may perform differently.
- **Fixed hyperparameters.** The exploration parameter $c = \sqrt{2}$ is the standard UCB1 value. Tuning c for educational settings (e.g., higher c for uncertain learners) is an open question.
- **Category-level granularity.** We select at the category level with 6 categories. Finer-grained selection (individual questions or skills) and larger numbers of categories would test scalability.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: All claims are supported by simulation results presented in Section 4. We clearly state the simulation-based nature of the study.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Limitations are discussed in Section 5 and Section S3 of the supplementary material, including the use of simulated (not real) students and simplified learning dynamics.

3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (correct) proof?

   Answer: [NA]

   Justification: This is an empirical simulation study. We reference the theoretical foundations of UCB1 from prior work [7].

4. **Experimental Result Reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

   Answer: [Yes]

   Justification: All simulation parameters are reported in Section 3, the random seed is fixed at 42, and we provide complete source code.

5. **Open Access to Data and Code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

   Answer: [Yes]

   Justification: The complete source code, experiment runner, and result data will be released on GitHub.

6. **Experimental Setting/Details**

   Question: Does the paper specify all the training and test details necessary to understand the results?

   Answer: [Yes]

   Justification: All parameters (learning rate, decay rate, exploration constant, population size, number of questions) are specified in Table 1 and Section 3.

7. **Experiment Statistical Significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: We report standard deviations across the student population and analyze results across three independent runs with different seeds.

8. **Experiments Compute Resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiment?

   Answer: [Yes]

Justification: Computational requirements are minimal (standard CPU, under 1 minute per run) and are described in the supplementary material.

9. **Code of Ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

   Answer: [Yes]

   Justification: This simulation study uses no real student data and raises no ethical concerns.

10. **Broader Impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work?

    Answer: [Yes]

    Justification: We discuss the potential for more efficient personalized learning and note the need for validation with real students before deployment.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models?

    Answer: [NA]

    Justification: The released code is a simulation framework with no potential for misuse.

12. **Licenses for Existing Assets**

    Question: Are the creators or original owners of assets used in the paper properly credited?

    Answer: [Yes]

    Justification: We cite the Gymnasium framework [13] and all algorithmic foundations.

13. **New Assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The simulation framework includes documentation, a README, and CLI help.

14. **Crowdsourcing and Research with Human Subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants?

    Answer: [NA]

    Justification: No human subjects were involved; all students are simulated.

15. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs in the research process?

    Answer: [Yes]

    Justification: An LLM (Claude) was used to assist with paper writing and code documentation. No LLMs were used in the experimental methodology or data generation.